Caesar Cipher

The Caesar cipher is arguably the most well known cipher in existence. This cipher is one of many considered to be a substitution cipher, where each letter in the original message (also known as the plaintext) is replaced with a letter corresponding to a certain number of letters shifted up or down in the alphabet (resulting in the ciphertext). This method was created and implemented by none other than Julius Caesar, who used this cipher to send messages to his troops. Since the Caesar cipher is bound to whatever language you're using it with, it is extremely vulnerable to brute force attacks. For instance, in the English language, you can only shift the language 27 times before you end up back at the beginning. This means the keyspace (number of possible encryptions) for the Caesar cipher is 27. As such, a brute force method would have the solution within 27 attempts.

ATBash Cipher

The ATBash cipher is considered to be one of the simplest ciphers available. This Cipher works by simply reversing the alphabet (ie. A->Z, B->Y, etc). As such, the ATBash cipher is a simple substitution cipher. Since this Cipher is well known, it is considered extremely easy to crack. This cipher is quite old, and as such it is no longer known who made the cipher or when. It is known, however, that its intended initial use was with the Hebrew alphabet. This is also where the ATBash Cipher derives its name from.

Vigenère Cipher

The Vigenere cipher is a more complex cipher as opposed to ATBash or Caesar. While this cipher was originally described by Giovan Battista Bellaso, it's more often accredited to 16th century mathematician Blaise de Vigenère (from whom the ciphers name later came from). Vigenère's goal when creating this cipher was to add a twist on the Caesar Cipher, by making it more impervious to frequency analysis. He accomplished this by implementing a word or phrase that acted as a key. You would loop through this phrase, with each letter correlating to a different letter alphabet shift. This is considerably safer than the Caesar cipher, and at the time it was almost impossible to crack without that passphrase. In the modern day, however, the Vigenère cipher has become vulnerable to frequency analysis and brute force attacks.

Four Square Cipher

The Four Square cipher is what's known as a polygraphic substitution cipher, encrypting pairs of letters (known as digraphs), as opposed to individual letters. In addition, it separates itself from the other ciphers by utilizing two different keys. Created by Felix Delastellein the 1800s, this form of cipher is considered safer than simple substitution ciphers, as it is comparatively more resilient to frequency analysis. This cipher requires the use of 4 5x5 matrices (often leaving out the letter Q or J). In the top left and bottom right are normal matrices, corresponding to the alphabet. In the top right and bottom left however, are 5x5 matrices with a key moved to the front of the alphabet. For instance, if the key was "Mouse", the matrix will begin by spelling out the word mouse. It will then proceed to finish the rest of the alphabet, skipping letters already used by the key. By lining up these 5x5 matrices and arranging them in a 2x2, you are able to produce the cipher text. This is done by imagining a hypothetical 5th square, with the digraph letters being the top left corner and the bottom right. By taking the opposite corners (top right and bottom left) and repeating this process, you produce the cipher text. In comparison to the other classical ciphers, the Four Square cipher is considered one of the most secure and time consuming. However, much like all the ciphers present, are not considered safe in the modern day. As it is well known technique, a computer would be able to crack it quickly.

Playfair Cipher

Created in the 1800s by Charles Wheatstone, the Playfair cipher is one of the first diagram substitution ciphers created. Not unlike the Four Square cipher, Playfair encrypted pairs of letters at a time (called bigrams). Unlike Four Square, however, Playfair required only one key and one 5x5 matrix. As it was simpler to use, the Playfair cipher was preferred over Four Square. Once the key was moved to the front of the matrix (such as with Four Square), three rules were applied to produce the ciphertext:

- If the letters of the bigram are in the same row, replace them with the letters to their right (wrapping around to the left when necessary).
- If the letters of the bigram are in the same column, replace them with the letters below (wrapping around to the top when necessary).
- If the letters form a rectangle (meaning the previous 2 rules don't apply), replace them with the letters on the opposite corners of the rectangle.

In order to crack this cipher, you would just need to reverse the process. This cipher was notable as it had seen use during both World War 1 and World War 2. It's worth noting that, at the time, it was possible to crack the cipher by simply guessing the words. As such the Playfair cipher was used to encrypt important messages, but not messages of the utmost importance.